Model 1 Stress Testing Report (Abishag)

1. Model Overview

Model 1 estimates daily energy consumption (kWh) using an XGBoost regression model based on DC and AC power inputs. The model is deployed via a FastAPI API and provides predictions for energy consumption based on provided input values.

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2. Acceptable Input Ranges

Variable	Range	Notes
dc_power	≥ 0 and ≤ 10,000	Must be non-negative and realistic
ac_power	≥ 0 and ≤ 10,000	Must be non-negative and realistic

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3. Edge Cases Tested

1. Normal values: dc_power = 500, ac_power = 450 → Output: 2563.98 kWh
2. Zero inputs: dc_power = 0, ac_power = 0 → Output: 0.0 kWh
3. Negative inputs: dc_power = -100, ac_power = 300 → Error: "Power values must be non-negative"
4. Extremely high values: dc_power = 9,000,000, ac_power = 300,000 → Error: "Power values exceed realistic maximum"
5. Missing input: dc_power = 500, ac_power = *blank* → Error: "All input fields are required"

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4. Observed Failure Modes (Before Correction)

- Previously, zero and negative inputs produced unrealistic outputs.
- Extremely high input values produced illogical predictions.
- Missing inputs were already handled correctly.

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5. Corrective Actions Implemented

- Added **input validation** for negative, zero, and extreme inputs.
- Added logic to return **0 kWh when both inputs are zero**.
- Set upper bounds for realistic DC and AC power inputs.
- Ensured clear error messages are returned for invalid or missing inputs.